

Density processing

1 Raw active density γ

2 Active-mask Helmholtz filter $\tilde{\gamma}$

3 Projection and passive bounds $\bar{\gamma}$

4 Material fields $\alpha(\bar{\gamma})$
SA and wall penalties

Forward turbulent Picard loop

8 Picard residual accepted?

7 SA transport $\tilde{\nu}$ and ν_t

6 Flow solve IPCS/SNES $\mathbf{u}, p$

5 Reciprocal wall distance $G, y(G)$

repeat until accepted

accepted

9 Final frozen state solve $\mathbf{s} = (\mathbf{u}, p)$

10 Frozen flow adjoint $\mathbf{s}^*$

11 Sensitivities dJ, dV with respect to $\bar{\gamma}$

12 Projection slope and transpose filter

13 MMA update active raw DOFs

Frozen sensitivity and optimizer update

left-to-right sensitivity path

next MMA iteration